

Study to Determine Comparability of PFAS Drinking Water Samples in Cadillac, Michigan

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Summary

The purpose of this study was for the Michigan PFAS Action Response Team (MPART) to determine the comparability of using three different drinking water sampling methods by the same person or team at the same location and as close to the same time as possible. This study was conducted because some Cyclopure test results taken by residents in the spring of 2025 differed in comparison to samples taken roughly two months later by the Michigan Department of Environment, Great Lakes, and Energy (EGLE)'s contractor, AECOM.

For this study, the team used test kits/bottles from three labs:

- 1) **EGLE Drinking Water Lab**
- 2) **Merit Labs (Merit)**
- 3) **Cyclopure**

Using these three labs, MPART determined whether the same person sampling at the same location at approximately the same time (back-to-back) would generate similar results.

Background

EGLE, through its contractor AECOM, conducted drinking water sampling as part of an investigation in the Cadillac area and as of September 2025 had collected over **100** PFAS samples using EPA Method 533. Statewide, AECOM has collected thousands of PFAS samples from outside home spigots following MPART protocols.

Since the spring of 2025, both the City of Cadillac and Haring Township have been responding to drinking water sampling requests from residents by offering Cyclopure test kits. Cyclopure has analyzed thousands of samples nationwide using their test kits. In utilizing Cyclopure test kits, the City of Cadillac uses City employees to collect samples and Haring Township uses volunteer samplers. Both agencies collect samples from kitchen sinks. Meanwhile, many Cadillac residents are also using Cyclopure test kits on their own, sampling from their kitchen sinks.

In April 2025, as a part of the state's ongoing investigation in the Cadillac area, EGLE resampled several homes that had exceedances of Michigan's Drinking Water Criteria from Cyclopure test kits. The results for four homes (identified in [Figure 1](#) as light blue circles/purple squares circled in dark blue) did not align well. Specifically, four homes had levels of PFOS and PFOA only (no other PFAS) at fairly similar levels, while EGLE's resample at the home was non-detect. This resulted in confusion by residents and concern about the validity of the drinking water samples. It is important to note that the four samples in question were taken on different dates and times, by different samplers (e.g., the homeowners), and at a different location (at indoor faucets instead of outside spigots). This study attempted to eliminate as many of these variables as possible.

The goal of the project was to conduct a side-by-side sampling effort at the four homes identified above, plus up to another 16 properties in the Cadillac area. The sampling process was designed to mimic what a Cyclopure test user typically does inside the home and what EGLE typically does outside the home. This sampling was not intended as a PFAS source investigation.

Procedure

MPART sent recruitment letters to all residents who had shared their Cyclopure test results which had exceedances of Michigan's Drinking Water standards. In addition, MPART sent recruitment letters to some properties with detections (but not exceeding criteria) in their Cyclopure test results, along with two properties with non-detect Cyclopure test results. EGLE's sampling contractor, AECOM, sampled the first 20 residents that responded. Residents/properties invited to participate in the study are identified on the map in [Figure 1](#). Letters offered free testing at the property, which was conducted accordingly:

- Before sampling:
 - AECOM had two samplers at each location, which are referred to as Sampler One and Sampler Two. Both samplers pre-labeled bottles for all three tests at each location (sink and spigot), with the address, date, and a unique lab ID number. Labeling was done with a PFAS-free permanent marker.
 - Both samplers followed principles in MPART's [General PFAS Sampling Guidance](#) and [Quick Reference Field Guide](#).
 - The AECOM samplers noted whether the water collected from the kitchen faucet was unfiltered, filtered, or softened. If filtered, samplers noted what kind of filter, specifically make and model, if available. AECOM samplers also noted the date of the sample and address, pre-labeling as much on the bottles as possible.
- Sampling occurred on September 9 and 10, 2025. Steps:
 - **For as many of the 20 properties as possible, one set of samples was collected from the kitchen faucet according to the sampling method used by most homeowners** (e.g. per the [Cyclopure test kit instructions](#)). This process, according to the Cyclopure instruction sheet, does **not** require running the water for two to three minutes.
 - Both samplers donned nitrile gloves.
 - AECOM Sampler One followed the instructions for a Cyclopure test kit. Upon collecting the Cyclopure test sample, Sampler One shut off the faucet.
 - AECOM Sampler Two placed the Cyclopure sample on a stand designed by Cyclopure for the approximately 20 to 30 minutes it takes the sample water to drain from the cup. Sampler Two noted the time of the Cyclopure sample on the chain of custody form.

- While AECOM Sampler Two waited on the Cyclopure sample to drain, EGLE Sampler One changed gloves if they touched any item other than a sample bottle and then collected a water sample from the faucet—without running the water for two to three minutes—using the EGLE lab bottles. Sampler Two noted the time the EGLE lab sample was taken on the chain of custody. Sampler One shut off the faucet.
 - Immediately after the EGLE sample was taken and the bottle top properly placed, AECOM Sampler One changed gloves if they touched any item other than a sample bottle and then collected a sample from the faucet—without running the water for two to three minutes—using the Merit bottles. Sampler Two noted the time the Merit lab sample was taken. Sampler One shut off the faucet.
 - AECOM Sampler One capped the Merit sample, making sure both samples were properly labeled and put into a cooler following normal EGLE sampling storage procedures.
 - EGLE Sampler Two processed the Cyclopure test per the instructions provided and gave the sample to the Cyclopure representative to store it in a cooler or box.
- **For as many of the 20 properties as possible, a second set of samples was taken from the *outside spigot* following EGLE’s sampling process** as per [EGLE’s Residential Well PFAS Sampling Guidance document](#). For this study the piping was flushed by running the water for exactly three minutes at a moderate volume. This was accomplished by Sampler One who had donned nitrile gloves. All flushed water was run into a bucket and the bucket emptied onto the ground at the sampling location. After the water line had been flushed for three minutes, EGLE Sampler Two turned off the faucet. As soon as possible afterwards:
- EGLE Sampler One donned new nitrile gloves and turned the faucet back on at low volume and filled the Cyclopure test kit per the instructions (but again, in this case, using water flushed for three minutes). Once the Cyclopure test was complete, EGLE Sampler One handed the test kit to EGLE Sampler Two who placed the test kit in a decontaminated five-gallon bucket with holes drilled in the lid to drain. Sampler Two also noted the time the sample was taken.
 - Immediately after the Cyclopure sample was taken, EGLE Sampler One donned fresh nitrile gloves if they touched any item other than a sample bottle and took a sample using the EGLE lab bottles. Once the bottle top was put on properly, they placed the bottles in the cooler. Sampler Two noted the time the samples were taken.
 - Immediately after the EGLE lab bottles were put into the cooler, EGLE Sampler One donned fresh nitrile gloves if they touched any item other than a sample bottle and took a sample using the Merit lab bottles. Once the bottle top was put on properly, they placed the bottles in the cooler. Sampler Two noted the time the sample was taken.
 - EGLE Sampler Two processed the Cyclopure test per the instructions provided and gave the sample to the Cyclopure representative to store it in a cooler or box.

This process mimicked what a Cyclopure test user typically does inside the home and what EGLE typically does outside the home.

- **For two selected properties that had detections**, the samplers conducted the above indoor/faucet sampling as per the instructions above (e.g., no flushing). In addition, the samplers **took a second set of samples at the indoor faucet after running water for 3 minutes** before the samples were taken, following all of the procedures listed.
- Samplers ensured that the appropriate field blanks and duplicates were taken from a selected property.
- At select properties, if residents were not available to let samplers inside the home, a set of samples was still taken from the outside spigot. Alternatively, if an outside spigot was not available, only an indoor sample was taken.
- At two properties, whole house filters had been recently installed by the resident. At the request of the resident, the samples were still collected from the inside and outside locations, despite being considered a “filtered” sample. Although the sample results would not necessarily reflect groundwater concentrations, the purpose of comparing sampling methods was still achieved.

Shipping

Cyclopure samples were driven by Cyclopure staff to the Cyclopure lab in Chicago. Consistent with EPA Method 533, EGLE and Merit samples were kept on ice and dropped off at the EGLE and Merit Laboratories in Lansing and East Lansing by AECOM.

Supplies Used

- Sample bottles/test kits provided by the **EGLE** and **Merit** Labs, and **Cyclopure**.
- Coolers for samples collected by EPA Method 533 (EGLE and Merit) to meet the cooling requirements of the EPA method.
- Cyclopure samples do not require cooling upon sampling, so a box or a cooler without ice was used for those samples.
- Field blanks and duplicates were collected in accordance with the EPA method and EGLE guidance.
- Nitrile gloves.
- Cyclopure stands for draining Cyclopure samples.
- Buckets (for running outside tap water and dumping water on-site as well as draining outdoor Cyclopure samples).

Results

Results with the addresses removed are shown in [Table 1](#). It is important to note the following:

- Each lab has its own format for presenting results. To make results easier to read, each participant received a summary table of all their results along with each individual lab report. If a participant had used a Cyclopure test in the spring of 2025, EGLE included those results in their side-by-side table, for comparison.
- Merit Laboratory and the EGLE Laboratory use the same EPA-approved method, which analyzes for 25 compounds, while Cyclopure uses their own method and analyzes for 55 PFAS compounds.
- The reporting limit (RL) is the smallest concentration of a PFAS analyte that a laboratory can reliably report. Concentrations below the RL are reported as not detectable (“ND”). In the case of the EGLE and Merit labs, the RL is 2 parts per trillion (ppt). Cyclopure has an RL of 1 ppt. Therefore, a Cyclopure result of 1.2 ppt, for example, would be reported as non-detect by both Merit and EGLE.
 - If a laboratory does not detect a specific PFAS chemical in a drinking water sample, it does not necessarily mean that PFAS was absent from the sample altogether. Rather, it could mean that the concentration of the PFAS chemical was too low to be measured accurately.
- Because labs are measuring these compounds at very small amounts, (i.e., parts per trillion) MPART considers results that are within a few parts per trillion of each other to be reasonably similar.
- Cyclopure’s instructions do not require flushing water before sampling. Both Merit and EGLE sampling protocols require flushing for three minutes. Homeowners that follow Cyclopure’s instructions ([Figure 2](#)) and take their samples inside the home at the kitchen faucet are likely sampling water that represents what the homeowner is typically drinking. The EGLE and Merit sampling method—flushing the water for three minutes and taking the sample from an outside spigot—more likely reflects the groundwater at that home.

Conclusions

Overall, the EGLE, Merit and Cyclopure test results taken during this sampling effort showed good comparability between the three different lab results. Results between the three laboratories were typically within a few parts per trillion of one another.

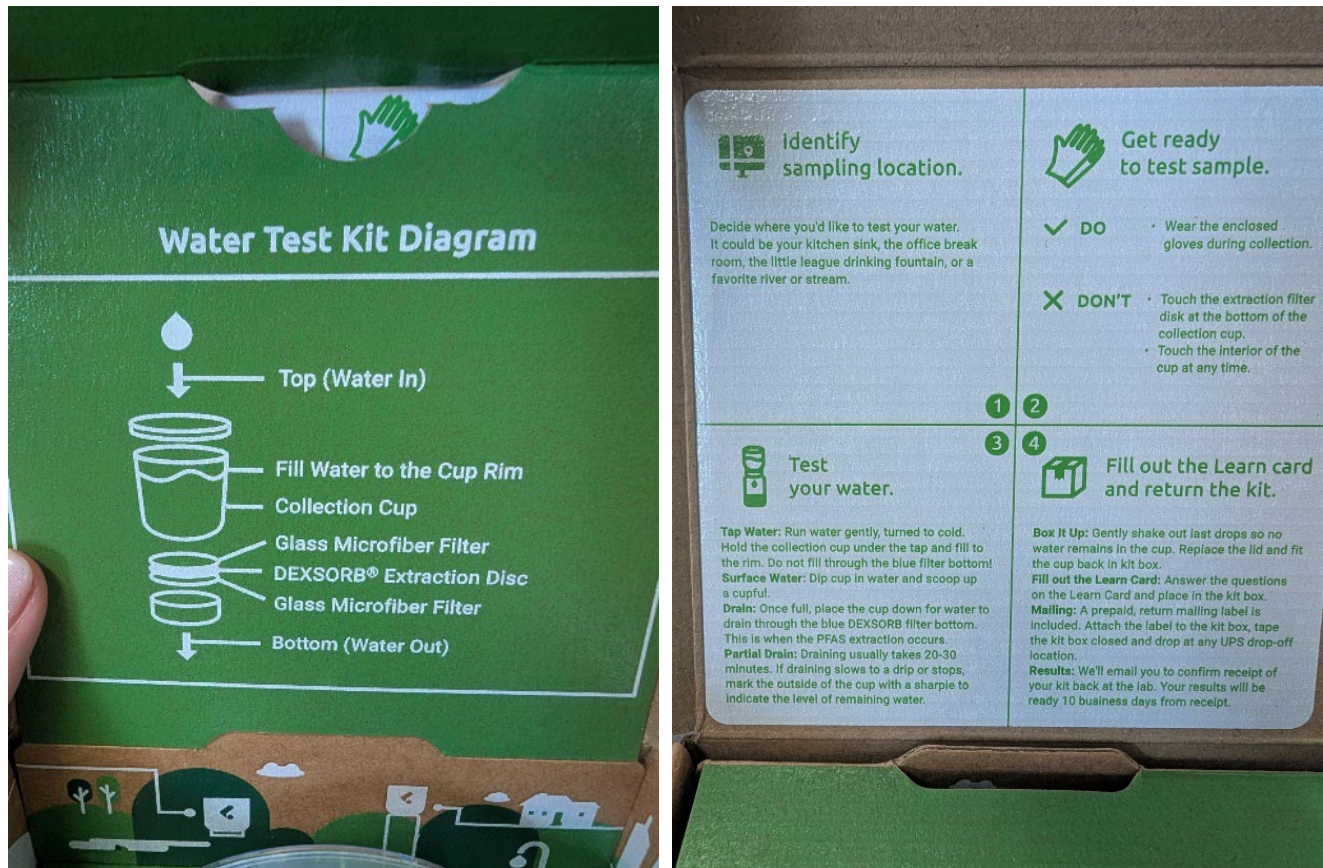
In general, MPART did not observe any notable differences between the indoor and outdoor samples or between the flushed and un-flushed samples. One exception to this was a property which had higher PFOA concentrations for all three methods in the outdoor sample compared to the indoor sample. However, for this same house, the other detected PFAS were found at similar concentrations when comparing the indoor and outdoor samples (see Address 8 in [Table 1](#)).

Of the four homes that had elevated levels of PFOS and PFOA in the Cyclopure test kits in the spring of 2025 that were then re-sampled by EGLE, two residents agreed to participate in the comparability study. Both homes were non-detect inside and outside the home for all three labs. Note that one of the homes that had elevated PFOS and PFOA in the spring from a Cyclopure test kit is near a field where biosolids had been applied. Since all subsequent tests were non-detect, the data suggests that the home's drinking water well is not being impacted by biosolids.

In conclusion, this study achieved good comparability between the three different labs and the outdoor and indoor sampling locations.

Figure 1: Properties that were sampled in the comparability study and additional properties that were invited to participate



Figure 2: Cyclopure test instructions**Figure 3: Michigan PFAS drinking water criteria**

Specific PFAS	Drinking water criteria (ppt*)
PFOA	8 ppt
PFOS	16 ppt
PFNA	6 ppt
PFHxS	51 ppt
PFBS	420 ppt
HFPO-DA (GenX)	370 ppt
PFHxA	400,000 ppt

*ppt = parts per trillion

Table 1: Drinking Water Comparability Study Results (addresses removed)

Cadillac, Michigan - September 9 and 10, 2025

Address 1

Analyte	Indoor EGLE	Indoor EGLE Dup	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND	ND
PFPeA	ND	2	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND	ND
Treatment comments:	Indoor – water softener				Outdoor – no treatment		

Bold = Detection of PFAS compound;**Red border** = Exceedance of MI Drinking Water Criteria

Address 2

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND
Treatment comments:	Indoor – no treatment			Outdoor – no treatment		

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 3

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – whole-house PFAS filter and water softener

Outdoor – whole-house PFAS filter and water softener

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 4

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – whole-house PFAS filter and water softener

Outdoor - whole-house PFAS filter

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 5

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	1.7	ND	ND	1.6
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	3	3	1.2	3	3	1.3
PFBS	3	2	2.3	3	3	3
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	1.4	ND	ND	1.6
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	4	4	3.3	4	3	3.8
PFHxS	6	5	5.1	5	4	5.8
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	10	6	6.3	9	6	6.8
PFOS	6	5	4.6	6	5	6
PFPeA	4	3	2.3	5	3	2.5
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND
Treatment comments:	Indoor – no treatment			Outdoor - no treatment		

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 6

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – no treatment

Outdoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 7

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	4	3	1	4	3	ND
PFBS	9	7	7.1	10	8	7
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	7	6	4.9	7	6	4.5
PFHxS	ND	ND	1.4	3	ND	1.7
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	4	2	2.1	4	3	2.7
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	7	5	3.9	7	5	3
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – non-PFAS filter and water softener

Outdoor – water softener

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 8

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	6	5	5.7	6	5	5
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	2	ND	2	2	ND	1.9
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	5	4	4.9	4	4	3.9
PFHxS	13	11	12.8	14	11	11.9
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	7	4	4.3	14	10	10.9
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	5	4	3.7	4	4	2.7
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND
Treatment comments:	Indoor – no treatment, bathroom sink			Outdoor – no treatment		

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 9

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – no treatment

Outdoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 10

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND
Treatment comments:	Indoor – Point-of-use PFAS filter			Outdoor – no treatment		

Bold = Detection of PFAS compound; **Red border** = Exceedance of MI Drinking Water Criteria

Address 11

Analyte	Indoor EGLE QC	Indoor EGLE	Indoor EGLE Dup	Indoor Merit	Indoor Cyclopure	Indoor (Flushed)* EGLE	Indoor (Flushed)* Merit	Indoor (Flushed)* Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFBA	5	5	5	4	1.6	5	4	1.7	5	4	1.2
PFBS	30	28	29	24	22.5	29	27	24.6	28	26	24.2
PFDA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFHpA	4	4	4	3	3.6	4	4	3.7	4	4	3.4
PFHpS	ND	ND	ND	ND	1.9	ND	ND	1.6	ND	ND	1.9
PFHxA	5	5	5	5	4.5	5	6	3.9	5	5	3.4
PFHxS	38	36	39	29	31.7	35	29	32.6	37	28	31
PFMBA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFOA	36	34	35	22	21.4	35	30	21.4	35	24	20.3
PFOS	20	19	18	14	12.9	18	15	12.4	18	14	12.1
PFPeA	4	4	4	3	1.5	4	3	1.7	4	3	1.6
PFPeS	2	2	2	ND	1.8	3	ND	1.8	2	ND	1.7
PFUnA	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – No treatment

Outdoor – no treatment

* Flushed for three (3) minutes

Bold = Detection of PFAS compound; Red border = Exceedance of MI Drinking Water Criteria

Address 12

Analyte	Non-outdoor EGLE	Non-outdoor Merit	Non-outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND
4:2FTS	ND	ND	ND
6:2FTS	ND	ND	ND
8:2FTS	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND
ADONA	ND	ND	ND
HFPO-DA	ND	ND	ND
NFDHA	ND	ND	ND
PFBA	5	4	1.8
PFBS	9	8	7.6
PFDA	ND	ND	ND
PFDaA	ND	ND	ND
PFEESA	ND	ND	ND
PFHpA	ND	ND	1.2
PFHpS	ND	ND	ND
PFHxA	3	4	3.3
PFHxS	2	ND	1.5
PFMBA	ND	ND	ND
PFMPA	ND	ND	ND
PFNA	2	ND	1.5
PFOA	5	3	2.5
PFOS	12	9	8.6
PFPeA	4	2	1.6
PFPeS	ND	ND	ND
PFUnA	ND	ND	ND

Treatment comments:

No outside spigot - sampled untreated spigot located in bathroom in lieu of outside spigot

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 13

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – Point-of-use PFAS filter

Outdoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 14

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – no treatment

Outdoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 15

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND
4:2FTS	ND	ND	ND
6:2FTS	ND	ND	ND
8:2FTS	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND
ADONA	ND	ND	ND
HFPO-DA	ND	ND	ND
NFDHA	ND	ND	ND
PFBA	ND	ND	1.3
PFBS	ND	ND	ND
PFDA	ND	ND	ND
PFDaA	ND	ND	ND
PFEESA	ND	ND	ND
PFHpA	ND	ND	ND
PFHpS	ND	ND	ND
PFHxA	ND	ND	ND
PFHxS	ND	ND	ND
PFMBA	ND	ND	ND
PFMPA	ND	ND	ND
PFNA	ND	ND	ND
PFOA	ND	ND	ND
PFOS	ND	ND	ND
PFPeA	ND	ND	ND
PFPeS	ND	ND	ND
PFUnA	ND	ND	ND

Treatment comments:

Indoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 16

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	ND
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	ND	ND	ND	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	ND	ND	ND	ND	ND	ND
PFOS	ND	ND	ND	ND	ND	ND
PFPeA	ND	ND	ND	ND	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – no treatment

Outdoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 17

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND
4:2FTS	ND	ND	ND
6:2FTS	ND	ND	ND
8:2FTS	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND
ADONA	ND	ND	ND
HFPO-DA	ND	ND	ND
NFDHA	ND	ND	ND
PFBA	ND	ND	ND
PFBS	ND	ND	ND
PFDA	ND	ND	ND
PFDaA	ND	ND	ND
PFEESA	ND	ND	ND
PFHpA	ND	ND	ND
PFHpS	ND	ND	ND
PFHxA	ND	ND	ND
PFHxS	ND	ND	ND
PFMBA	ND	ND	ND
PFMPA	ND	ND	ND
PFNA	ND	ND	ND
PFOA	ND	ND	ND
PFOS	ND	ND	ND
PFPeA	ND	ND	ND
PFPeS	ND	ND	ND
PFUnA	ND	ND	ND

Treatment comments:

Indoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 18

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	5	4	1.1	5	4	ND
PFBS	6	6	5.3	6	5	4.9
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	ND	ND	ND	ND	ND	ND
PFHxS	2	ND	2	3	ND	1.5
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	5	4	4.1	5	4	3.5
PFOS	7	4	4.9	7	5	5.6
PFPeA	2	ND	1.2	2	ND	ND
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – water softener turned off

Outdoor – no treatment

Bold = Detection of PFAS compound;

Red border = Exceedance of MI Drinking Water Criteria

Address 19

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND
PFBA	ND	ND	ND	ND	ND	ND
PFBS	ND	ND	ND	ND	ND	1.6
PFDA	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND
PFHpA	ND	ND	ND	ND	ND	ND
PFHpS	ND	ND	ND	ND	ND	ND
PFHxA	2	2	2.1	2	2	2.6
PFHxS	ND	ND	1.2	ND	ND	ND
PFMBA	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND
PFOA	4	3	2.7	4	3	3.3
PFOS	5	4	5.1	5	4	5.5
PFPeA	3	ND	1.4	3	ND	1.4
PFPeS	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – no treatment; flushed by homeowner right before sampling

Outdoor – water softener on outside spigot

Bold = Detection of PFAS compound; **Red border** = Exceedance of MI Drinking Water Criteria

Address 20

Analyte	Indoor EGLE	Indoor Merit	Indoor Cyclopure	Indoor (Flushed)* EGLE	Indoor (Flushed)* Merit	Indoor (Flushed)* Cyclopure	Outdoor EGLE	Outdoor Merit	Outdoor Cyclopure
11Cl-PF3OUdS	ND	ND	ND	ND	ND	ND	ND	ND	ND
4:2FTS	ND	ND	ND	ND	ND	ND	ND	ND	ND
6:2FTS	ND	ND	ND	ND	ND	ND	ND	ND	ND
8:2FTS	ND	ND	ND	ND	ND	ND	ND	ND	ND
9Cl-PF3ONS	ND	ND	ND	ND	ND	ND	ND	ND	ND
ADONA	ND	ND	ND	ND	ND	ND	ND	ND	ND
HFPO-DA	ND	ND	ND	ND	ND	ND	ND	ND	ND
NFDHA	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFBA	2	ND	ND	2	ND	ND	ND	ND	ND
PFBS	6	5	5.8	7	6	5.2	7	6	5.9
PFDA	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFDaA	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFEESA	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFHpA	2	ND	1.6	2	ND	1.3	2	ND	1.6
PFHpS	ND	ND	ND	ND	ND	ND	7.1	ND	ND
PFHxA	5	4	4.3	5	4	3.1	4	4	4
PFHxS	45	33	41.6	44	33	34.3	46	36	43
PFMBA	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFMPA	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFNA	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFOA	10	6	7.2	10	7	5.6	10	6	7.1
PFOS	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFPeA	3	2	2.4	3	2	1.7	3	2	1.8
PFPeS	ND	ND	ND	ND	ND	ND	ND	ND	ND
PFUnA	ND	ND	ND	ND	ND	ND	ND	ND	ND

Treatment comments:

Indoor – unfiltered, water softener without salt

Outdoor – no treatment

* Flushed for three (3) minutes

Bold = Detection of PFAS compound; **Red border** = Exceedance of MI Drinking Water Criteria